To: Board of Directors
From: Pierre Marson, Head of ALCO
Re: Crédit Géneral, S.A.
Date: June 13, 1995

I. Executive Summary

Crédit Géneral, S.A. agreed to a deal with a German corporate client that had recently collected on a £1 billion receivable and wished to convert it into Deutschemarks. Crédit Géneral, S.A. agreed to do the trade on a principal basis at a reasonable price, quoting and guaranteeing a price to the client. The deal would enhance Crédit Géneral, S.A.'s stature in the sterling market, which was considerably more illiquid compared to major currencies. Crédit Géneral, S.A.'s foreign exchange group concluded that the yard sterling could be laid off relatively easily if all channels were hit as rapidly and as close to simultaneously as possible.

The information gathered by the traders suggested that the sterling market would continue the relatively calm pattern it had exhibited during the last week. Expected market volatility, as implied by the prices of one-month sterling put and call options, was slightly lower than in recent days. Moreover, Crédit Géneral's traders had received indications of interest from a large U.S. bank to buy as much as several hundred million pounds. They also had received indications of interest from many other market makers to buy in smaller sizes.

Upon the execution of the deal, the sterling market suddenly "evaporated." The price of sterling fell rapidly, as did liquidity. The Crédit Géneral traders quickly laid off what they could on the U.S. bank – £250 million – at a 3.5 pfenning loss. By late afternoon, they managed to lay off an additional £50 million, leaving the bank with its current exposure of **£700 million**. Sterling, quoted at 2.2406/2.2413 prior to the crash, now is quoted on Reuters screens at a 10 pip spread: 2.2013/2.2023.

The objective of the proposed trades is to materially reduce overnight tail risk while avoiding further market disruption. My recommendation is to liquidate £200 million and cross-currency hedge the remaining £500 million. The strategy balances **nominal exposure reduction** and **tail-risk containment,** and the split optimizes risk reduction and execution cost.

II. Background: VaR Calculation

Over the last 3 years, the worst one-day sterling move was **-1.3%**, and the worst ten-day sterling move was **-4.1%**. Hence, the one- and ten-day value-at-risk was calculated to be **DM 4.4 million** and **DM 13.8 million**, respectively, based on a maximum allowed position of **£150 million**.

Using the 2.2356 DM/ £ exchange rate, the maximum allowed position was **£150 million x 2.2356 DM / £ = <u>DM 335.34 million</u>.**

I. One-day VaR at 99% confidence interval (based on historical worst moves):

DM 335.34 million x -1.3% ≈ **<u>DM 4.4 million</u>**

II. Ten-day VaR at 99% confidence interval (based on historical worst moves):

DM 335.34 million x -4.1% ≈ **<u>DM 13.8 million</u>**

The VaR was computed for a **£150 million** position over one day based on historical price changes under the assumption of normal liquidity.

The realized loss of **DM 10.5 million** (2.4x one-day VaR) stems from the same-day forced execution on **£300 million**. The unrealized loss of **DM 24 million** (5.5x one-day VaR) reflects marking a **£700 million** overnight position to a stressed market price.

The total loss of **DM 34.51 million** is **<u>7.8x the one-day VaR</u>**.

The jarring difference between the estimates and the true loss is explained by multiple factors:

1. The VaR position size and the true exposure grossly varied in size. Scaling the VaR reflects an estimated loss of (£700 million x 2.2356 DM / £) x -1.3% ≈ **DM 20.3 million**, effectively closing the gap.
2. VaR measures price volatility (under the assumption of normal liquidity) rather than execution risk. Here, the losses incurred by Crédit Géneral, S.A. are largely the result of a liquidity event.
3. VaR is not cap; it is a quantile metric.

The VaR calculation was <u>not</u> grossly inaccurate. It performed as expected given its assumptions, but did not capture liquidity and execution risk.

III. Evaluation of Risk-Reduction Strategies & Recommendation

Below is a list of possible action plans to address the **£700 million** exposure:

1. Immediate spot liquidation: Sell all (some) of the £700 million exposure directly in the spot market.
2. Partial liquidation, holding the remainder: Reduce exposure without a full fire sale.
3. Purchase of £/DM put options: Cap downside risk while retaining upside.
4. Cross-currency hedging using correlated currencies: Use a liquid proxy (e.g., $/DM) to reduce effective sterling risk.
5. No immediate action: Maintain the position overnight in anticipation of a rebound.

Given the illiquidity and volatility of the sterling market, both immediate spot liquidation and purchase of options is costly. Leaving the **£700 million** as an overnight position exposes the bank to unreasonable tail risk if the sterling loses its value more.

The current market conditions, combined with a need for a prompt response, highlight **partial liquidation** and **cross-currency hedging** as the only viable options. The goal is to pick the most effective strategy for maximum tail risk reduction per unit of cost.

Partial liquidation: The strategy would immediately reduce nominal sterling exposure, avoid correlation or future risks, and improve the position size. However, it would cause a high impact in the illiquid sterling market, lock in losses at the lower prices, and may signal distress that can further scare the market.

Cross-currency hedging: The strategy would immediately reduce tail risk, avoid further disruptions in the sterling market, preserve upside if sterling rebounds. It is also reversible if conditions change. However, due to a non-one correlation, it would result in an imperfect hedge, which may also deteriorate in stress. Additionally, it does not address governance optics regarding the oversized exposure.

Combination of partial liquidation and cross-currency hedging: The strategy balances **nominal exposure reduction** and **tail-risk containment.** It would limit market impact by liquidating within the limits of what the market can absorb and would reduce reliance on correlation.

My proposed solution is to combine partial liquidation and cross-currency hedging in our response to the day's events. To decide on an effective split, we must consider how much sterling we can sell without worsening market conditions. Given that liquidity failed to improve during the course of the afternoon, further liquidation should be modest.

IV. Proposed Trades and Expected Outcomes

Inputs	Values
Total sterling exposure (£m)	700
Spot rate DM/£	2.2356
Daily vol DM/£ (σ_X)	0.0052
Daily vol DM/$ (σ_Y)	0.0074
Correlation (ρ)	0.3
z-score (99%)	2.33

Liquidation (£ M)	Remaining (£ M)	DM Exposure (X)	Unhedged VaR	Hedge Ratio	Hedge Size Y (DM)	Portfolio σ (DM)	Hedged VaR (DM)
100	600	1341.36	16.25	-0.21	-282.77	6.65	15.50
200	500	1117.80	13.54	-0.21	-235.64	5.54	12.92
300	400	894.24	10.83	-0.21	-188.52	4.44	10.34

The proposed trade is to sell £200 million sterling spot and cross-currency ($/DM) hedge £500 million (using minimum-variance hedge ratio).

Based on the sensitivity analysis above across liquidation amounts of £100–£300 million, the most effective split is to liquidate £200 million and hedge the remaining £500 million. The hedged VaR for £200 million represents the knee in the curve, suggesting that further liquidation delivers diminishing incremental risk reduction relative to the expected increase in execution costs in the illiquid sterling market.

Finally, please note that the sensitivity analysis was conducted using a correlation of 0.3, making the recommended strategy more conservative. Using higher recent correlations would imply greater hedge effectiveness.